

Lab 2 - Technical Communication

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1. The SNOW-V cipher is used for encrypting voice communications in 4G/5G networks. In Algorithm 1, we define the secret key and the initialization vector loading phase of the SNOW-V stream cipher.

Algorithm 1 The key and IV setup phase

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1: procedure KEYIV_SETUP( $K, IV, \text{is\_aead\_mode}$ )
2:    $(a_{15}, a_{14}, \dots, a_8) \leftarrow (k_7, k_6, \dots, k_0)$   $\triangleright K = (k_7, k_6, \dots, k_0)$ 
3:    $(a_7, a_6, \dots, a_0) \leftarrow (iv_7, iv_6, \dots, iv_0)$   $\triangleright IV = (iv_7, iv_6, \dots, iv_0)$ 
4:    $(b_{15}, b_{14}, \dots, b_8) \leftarrow (k_{15}, k_{14}, \dots, k_8)$ 
5:    $(b_7, b_6, \dots, b_0) \leftarrow (0, 0, \dots, 0)$ 
6:    $(R_1, R_2, R_3) \leftarrow 0, 0, 0$ 
7:   for  $t = 1 \dots 16$  do
8:      $T_1 \leftarrow (b_{15}, b_{14}, \dots, b_8)$ 
9:      $z \leftarrow (R_1 \boxplus_{32} T_1) \oplus R_2$ 
10:    FSM_UPDATE()
11:    LFSR_UPDATE()
12:     $(a_{15}, a_{14}, \dots, a_8) \leftarrow (a_{15}, a_{14}, \dots, a_8) \oplus z$ 
13:    if  $t = 15$  then
14:       $R_1 \leftarrow R_1 \oplus (k_7, k_6, \dots, k_0)$ 
15:    end if
16:    if  $t = 16$  then
17:       $R_1 \leftarrow R_1 \oplus (k_{15}, k_{14}, \dots, k_8)$ 
18:    end if
19:  end for
20: end procedure
```

2. In ques2.tex file

1 Formatting Tables

Write L^AT_EX code to generate the following tables:

1. Table 1

Category	Subcategory 1	Subcategory 2	Details
Algebra	Linear Equations	Quadratic Equations	Basic Algebra Concepts
Calculus	Differentiation	Integration	Fundamental Calculus
Geometry	Circles	Triangles	Euclidean Geometry
Programming	Variables	Control Structures	Basics of Coding

Table 1: Simple L^AT_EX table with text and strings

2. Table 2

Topic		
Subject	Description	Key Concept
Algebra	Linear Equations	Slope and Intercept
	Quadratic Equations	Parabolas and Roots
Calculus	Differentiation	Derivatives and Slopes
	Integration	Area Under the Curve
Geometry	Euclidean Geometry	Angles and Triangles

Table 2: Mathematical Topics and Their Key Concepts

3. Table 3

Category	Values		Total
	Math Symbol	Numeric	
Algebra	$\alpha + \beta$	10	$10 + 7 = 17$
	$x^2 + y^2$	7	
Calculus	$\int_a^b f(x) dx$	15	$15 + 9 = 24$
	$\lim_{x \rightarrow 0} \frac{\sin x}{x}$	9	
Geometry	πr^2	20	$20 + 25 = 45$
	$\frac{1}{2}bh$	25	

Table 3: Incorporating math symbols

2 Bibliography and Referencing

2.1 Algebra and Its Applications

Algebra forms the foundation of many mathematical theories and applications. As seen in Table 1, algebraic topics like Linear Equations and Quadratic Equations, discussed in *Linear Algebra and Its Applications* by Gilbert Strang [3], are essential in understanding basic algebra concepts.

Furthermore, Table 2 elaborates on the key concepts associated with these algebraic topics, such as “Slope and Intercept” for Linear Equations, which are crucial for solving and graphing linear relationships.

2.2 Calculus in Modern Engineering

Calculus is a critical area of study, particularly for understanding change and motion in dynamic systems. Table 1 outlines the fundamental topics in calculus, such as Differentiation and Integration, as detailed in *The Calculus Lifesaver* by Adrian Banner [1].

In Table 2, we further explore these concepts, where “Derivatives and Slopes” and “Area Under the Curve” are identified as key concepts in understanding the applications of calculus in various fields, including physics and engineering.

2.3 Geometry in Theoretical and Applied Mathematics

Geometry, as detailed in Table 1, includes topics like Circles, Triangles, and Euclidean Geometry, which are foundational to understanding spatial relationships and properties of shapes.

The importance of geometry is further illustrated in Table 3, where mathematical symbols and numeric values are used to represent geometric concepts. For instance, the area of a circle, represented by πr^2 , and the area of a triangle, represented by $\frac{1}{2}bh$, highlight the practical applications of these geometric principles in fields such as architecture and engineering. These principles are also critical in computer graphics and vision, as explored in research articles such as “Deep Residual Learning for Image Recognition” by He et al. [2], and modern approaches like “Attention Is All You Need” by Vaswani et al. [4], which employ geometric transformations and spatial understanding in their algorithms.

References

- [1] Adrian Banner. *The Calculus Lifesaver: All the Tools You Need to Excel at Calculus*. Princeton University Press, Princeton, NJ, 2007.
- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [3] Gilbert Strang. *Linear Algebra and Its Applications*. Cengage Learning, Boston, MA, 4 edition, 2016.
- [4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS)*, pages 6000–6010. Curran Associates Inc., 2017.